SQL Analysis Report

Project: E-Commerce Sales Analysis

Query 1 — Total Revenue

Business Question: What is the total revenue generated by the platform?

```
SELECT
SUM(payment_ value) AS total_ revenue
FROM payments;
```

total_revenue
16008872.12

Insight: The total revenue generated from all orders on the platform is approximately 16 million. This metric provides a high-level view of the marketplace's financial performance.

Query 2 — Total Orders

Business Question: How many orders were placed on the platform?

```
SELECT
COUNT(DISTINCT order _id) AS total_ orders
FROM orders;
```

Result Grid | Filter Rows:

total_orders
99441

Insight: A total of 99,441 orders were placed by customers. This indicates the scale of transaction activity on the platform.

Query 3 — Monthly Sales Trend

Business Question: How do order volumes change over time?

```
SELECT
YEAR(order_ purchase_ timestamp) AS year,
MONTH(order_ purchase_ timestamp) AS month,
COUNT(order_ id) AS total_ orders
```

FROM orders

GROUP BY year, month

ORDER BY year, month;

year	month	total_orders
2016	9	4
2016	10	324
2016	12	1
2017	1	800
2017	2	1780

Result 28

year	month	total_orders
2017	3	2682
2017	4	2404
2017	5	3700
2017	6	3245
2017	7	4026

Result 28

year	month	total_orders
2017	8	4331
2017	9	4285
2017	10	4631
2017	11	7544
2017	12	5673

Result 28

year	month	total_orders
2018	1	7269
2018	2	6728
2018	3	7211
2018	4	6939
2018	5	6873

Result 28

year	month	total_orders
2018	6	6167
2018	7	6292
2018	8	6512
2018	9	16
2018	10	4

Result 28

Insight: Order volume increases steadily over time, with certain months showing higher activity. These peaks may correspond to seasonal sales events or promotions.

Query 4 — Top Product Categories by Revenue

Business Question: Which product categories generate the highest revenue?

SELECT

p. product_ category_ name,

SUM(oi. price) AS revenue

FROM order_ items oi

JOIN products p

ON oi . product_ id = p. product_ id

GROUP BY p. product_ category_ name

ORDER BY revenue DESC

LIMIT 10;

Insight: The top-performing categories contribute significantly to overall revenue. These categories should be prioritized in marketing and inventory planning.

product_category_name	revenue
beleza_saude	1258681.34
relogios_presentes	1205005.68
cama_mesa_banho	1036988.68
esporte_lazer	988048.97
informatica_acessorios	911954.32
moveis_decoracao	729762.49
cool_stuff	635290.85
utilidades_domesticas	632248.66
automotivo	592720.11
ferramentas_jardim	485256.46

Query 5 — Sales by State

Business Question: Which states generate the highest sales revenue?

```
SELECT
c. customer_ state,
SUM(pay. payment_ value) AS revenue
FROM orders o
JOIN customers c
ON o. customer_ id = c. customer_ id
JOIN payments pay
ON o. order_ id = pay .order_ id
GROUP BY c. customer_ state
ORDER BY revenue DESC;
```

customer_state	revenue
SP	5998226.96
RJ	2144379.69
MG	1872257.26
RS	890898.54
PR	811156.38
SC	623086.43
BA	616645.82
DF	355141.08
GO	350092.31
ES	325967.55
PE	324850.44
CE	279464.03
PA	218295.85
MT	187029.29
MA	152523.02
PB	141545.72

customer_state	revenue
CE	279464.03
PA	218295.85
MT	187029.29
MA	152523.02
PB	141545.72
MS	137534.84
PI	108523.97
RN	102718.13
AL	96962.06
SE	75246.25
TO	61485.33
RO	46.20
AM	27966.93
AC	19680.62
AP	16262.80
RR	10064.62

Result 30

Insights: States such as SP, RJ, and MG generate the highest revenue, indicating that these regions are the primary customer markets. Focusing marketing campaigns and improving delivery efficiency in these states could further increase sales performance.

Query 6 — Average Customer Rating

Business Question: What is the average customer review score?

```
SELECT
AVG(review_ score) AS avg_ rating
FROM reviews;
```

	avg_rating
▶	4.0864

Insight: The average review score of 4.08 out of 5 suggests that most customers are satisfied with their purchases, reflecting positive overall customer experience.

Query 7 — Payment Method Usage

Business Question: Which payment methods are most frequently used?

```
SELECT
payment_ type,
COUNT(*) AS usage_ count
FROM payments
GROUP BY payment_ type
ORDER BY usage_ count DESC;
```

	payment_type	usage_count
▶	credit_card	76795
	boleto	19784
	voucher	5775
	debit_card	1529
	not_defined	3

Insight: Credit card payments dominate the platform with over 76k transactions, while boleto is the second most used payment method. This highlights the importance of maintaining reliable credit card payment infrastructure for customer convenience.